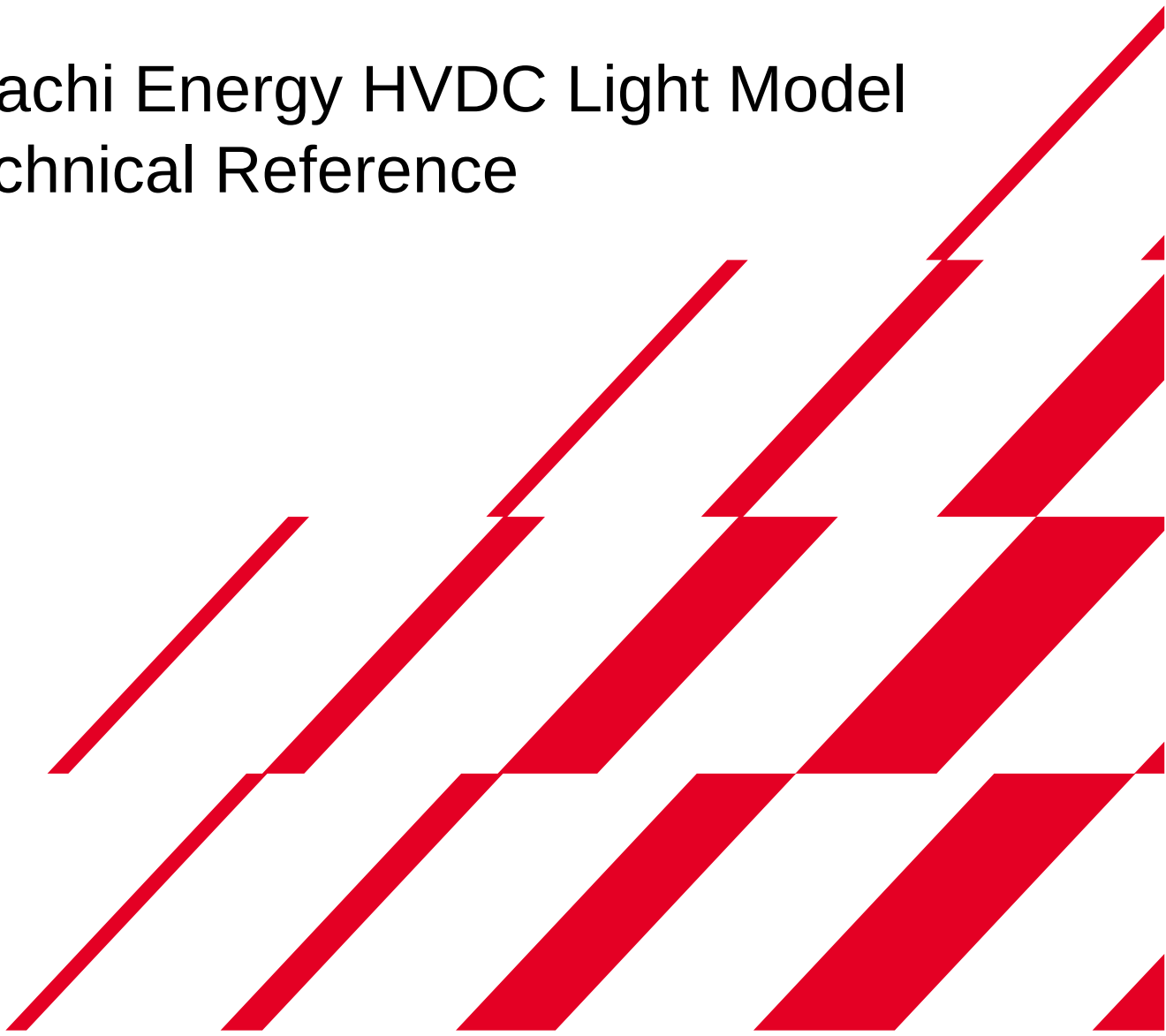
HITACHI
Inspire the Next

Hitachi Energy HVDC Light Model Technical Reference



Hitachi Energy

Table of Contents

1. Overview 3

2. Load flow representation 4

2.1 General settings 4
2.2 Converter 4
2.3 Converter transformer 5
2.4 Load flow settings in the PowerFactory converter model 6
2.5 Manual tap changer selection 6
2.6 Modulation index levels 6
2.7 PQ limitations 6

3. Dynamic model 8

3.1 General settings 8
3.2 Converter station frame 8
3.3 Input values 9
3.4 Parameter values 9
3.5 Output values 12
3.6 Error message 12
3.7 HVDC Light model Interface 12
3.8 Control system 15

4. References 17

1. Overview

This document gives instructions on how to set up the load flow, solve the load flow, run the dynamic simulation and operate the Hitachi Energy HVDC Light library model for PowerFactory. The model is limited to fundamental frequency AC quantities, described by phasors, and mean values of DC quantities.

The model consists of an end-to-end representation of the HVDC system, i.e., the electrical circuit between the Point of Common Coupling (PCC) buses, including converter stations, DC system and AC network equivalents. The model can be used for load flow calculations as well as dynamic simulation.

The converter stations and associated main circuit equipment, such as filters and power transformers, are represented using standard elements available in the PowerFactory program. Converter valves are modelled using the ElmVsc generic converter component. The DC side electrical representation is defined within PowerFactory by its standard components. The dynamics of the HVDC system is linked to the circuit components with user-written models.

2. Load flow representation

This section provides a detailed description of the load flow model including instructions on how to use it. Representation in the form that is described below is required for successful initialization of the dynamic model and the given rules must be followed to achieve proper function of the control system.

2.1 General settings

The HVDC model only represents positive sequence. In “Load Flow Calculation”, it is recommended to choose the option “Calculation Method: ACLoad Flow, balanced positive sequence”.

2.2 Converter

The HVDC Light converters are modelled using the standard PWM converter component ElmVsc within PowerFactory in the electrical circuit. The total cell capacitance of the converter arms is represented as an equivalent pole capacitor. The inductance of the arm reactors is represented in the PWM Converter component ElmVsc.

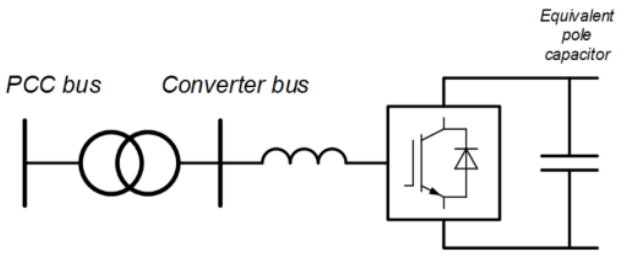


Figure 1: Overview of the converter main circuit.

The following parameters are valid for the HVDC Light converter, PowerFactory ElmVsc component.

Table 1: HVDC Light converter parameters.

Parameter Name	Parameter description	Unit	Coordination with dynamic model
Unom	Rated AC voltage	kV	Needs to be aligned with ‘UacBase’ value used in the dynamic model parameters
Unomdc	Rated DC voltage	kV	Needs to be aligned with ‘UdcNom’ value used in the dynamic model parameters
Snom	Rated power	MVA	Needs to be aligned with ‘Sb_Ctrl’ value used in the dynamic model parameters
Larm	Arm reactor inductance	mH	Needs to be aligned with ‘Lamr’ value used in the dynamic model parameters
Pcu	Series reactor Copper losses	kW	No corresponding dynamic control model parameter
Pnold	No Load losses	kW	No corresponding dynamic control model parameter
swtLossFactor	Switching loss factor	kW/A	No corresponding dynamic control model parameter
resLossFactor	Resistive loss factor	Ohm	No corresponding dynamic control model parameter

The equivalent pole capacitor is represented as two capacitors connected from terminal to the mid-point as shown in Figure 1.

Table 2: Equivalent pole capacitors.

Parameter Name	Parameter description	Unit	Coordination with dynamic model
ccap	Equivalent pole capacitor	uF	Used both in Load Flow and dynamic model. No specific alignment between the dynamic HVDC model and this parameter is needed

2.3 Converter transformer

The power transformer quantities to be specified are the MVA rating and the short circuit impedance. The converter bus and the PCC bus must be directly connected by a two-winding transformer.

The following parameters are valid, for modelling the converter transformer (ElmTr2) at each station.

Table 3: Converter transformers parameters.

Parameter name	Parameter description	Unit	Coordination with dynamic model
strn	Rated Power	MVA	Needs to be aligned with the ‘SbTrafo’ value used in the dynamic model parameters
frnom	Nominal frequency	Hz	Used both in Load Flow and dynamic model. No specific alignment between the dynamic HVDC model and this parameter is needed
utrn_h	Rated Voltage, HV side	kV	Needs to be aligned with ‘UlineNom’ value used in the dynamic model parameters
utrn_l	Rated Voltage, LV side	kV	Needs to be aligned with ‘UvalveNom’ value used in the dynamic model parameters
uktr	Reactance (ex) Short-Circuit Voltage (uk)	%	Needs to be aligned with ‘TrafoUk’ value used in the dynamic model parameters
pcutr	Copper losses	kW	Used both in Load Flow and dynamic model. No specific alignment between the dynamic HVDC model and this parameter is needed
dutap	Additional voltage per Tap	%	Used both in Load Flow and dynamic model. No specific alignment between the dynamic HVDC model and this parameter is needed
nntap0	Neutral position	-	Used both in Load Flow and dynamic model. No specific alignment between the dynamic HVDC model and this parameter is needed
ntpmn	Minimum position	-	Used both in Load Flow and dynamic model. No specific alignment between the dynamic HVDC model and this parameter is needed
ntpmx	Maximum position	-	Used both in Load Flow and dynamic model. No specific alignment between the dynamic HVDC model and this parameter is needed

2.4 Load flow settings in the PowerFactory converter model

The load flow settings of the PowerFactory converter model ElmVsc are described in the PowerFactory Technical References, PWM converter. Some clarifications are included here:

- One and only one converter shall be set in DC-voltage control (Vdc-Vac or Vdc-Q).
- The PowerFactory converter model ElmVsc have no optional selection of “controlled node” for the DC voltage in Load Flow. Consequently, in case the receiving converter (inverter) is in DC-voltage control mode the setpoint must be reduced to compensate for the DC-voltage drop to avoid an overvoltage at the sending end (rectifier). The load flow settings are independent of the settings for the dynamic simulation.
    - The operating mode in the dynamic simulation is set by the parameters described in Section 3.4.
    - The operating modes set in the PowerFactory converter model ElmVsc have only impact of the load flow.
    - Therefore, the operating mode in load flow can be selected different from the dynamic simulation, this can be used to select an operating mode that is favorable to improve the load flow solution.
- Independent of the operating mode settings in load flow and for dynamic simulations, the model will initiate properly to the operating point found by the load flow solution.

2.5 Manual tap changer selection

The tap changer position has to be set manually in PowerFactory, i.e. the tap position is set to a fixed value. The selection of fixed tap position can be done in different ways:

- If the HVDC Light reactive power is low (close to zero), a tap position corresponding to a winding voltage equal to the nominal winding voltage can be used.
- A detailed hand calculation.
- Iterative process:
    - Make a preliminary initialization and a short time dynamic simulation.
    - Monitor the modulation index.

Adjust the tap changer position if the modulation index is not within the recommended values given in 2.6.

2.6 Modulation index levels

Typical levels of modulation are:

- Maximum modulation index: 0.867.
- Minimum modulation index:0.820.

It shall be noted that the level of modulation index is more complex than that, but for RMS simulations this is a good enough representation.

2.7 PQ limitations

Within the operational limits tab of the ElmVsc component, the limits of active power are included as shown in Figure 2. It shall be noted that these limits refer to the terminals of the ElmVsc and therefore are compensated for the reactive power of the transformer to

comply with the PQ capability curve. Note that these limits are only valid for load flow. Dynamic limits are implemented separately in the dynamic model.

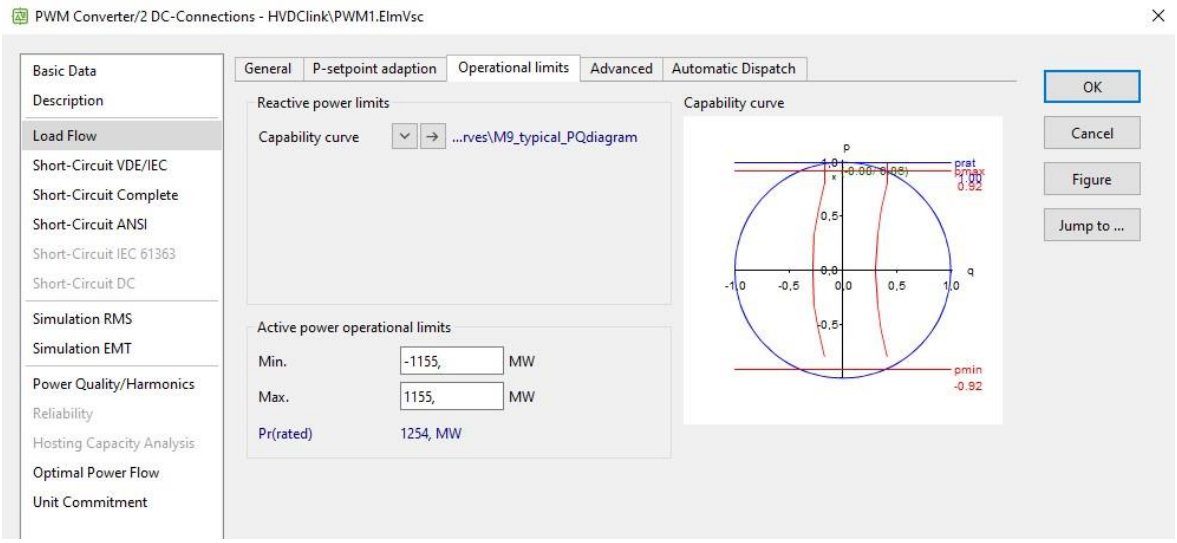


Figure 2: Load Flow operational limits.

3. Dynamic model

This section summarizes how the dynamic data of the model is defined and how the model is used. The input signals are measured from various points in the grid and passed-on to the control system. The control system processes the input signals and passes-on the output signals, which are the real and imaginary parts of the modulation index (i.e., Pmr and Pmi), to the block named PWM converter, which is then linked to the Voltage Source Converter (ElmVsc) used in the grid.

3.1 General settings

The HVDC model is design to perform electromechanical transient simulations and only represents positive sequence. In “Calculation of Initial Conditions” it is recommended to choose the settings “Simulation method: RMS values” and “Network representation: Balance, positive sequence”.

The model is suitable for time integration step in the range of 0,5 ms to 5 ms. The control system runs internally with an internal time step of 0,5 ms independent of the time step used in PowerFactory. For studies using weak AC networks, a time step in the range 0,5 ms to 2 ms is recommended. If Reference system “Reference: Nominal frequency” is chosen, a time step in the range 0,5 ms to 1 ms is recommended.

3.2 Converter station frame

The slots in the converter station frame are defined as in Table 4.

Table 4: Slots in the converter station frame.

Slot name	Type of element	Description
Time	ElmDsl	Provides the current time to the control block
Uac_ConverterBus_Meas	StaVmea	Measures the converter bus voltage
Uac_PCC_Meas	StaVmea	Measures the PCC bus voltage
Udc_meas_Pos	StaVmea	Measures the DC voltage at positive pole
Udc_meas_Neg	StaVmea	Measures the DC voltage at negative pole
Idc_meas_Pos	StaImea	Measures the DC current at positive pole
Idc_meas_Neg	StaImea	Measures the DC current at negative pole
I_converter_Meas	StaImea	Measures the phase current at converter bus Note: The reference direction is from converter towards the transformer
I_PCC_Meas	StaImea	Measures the phase current at line side of the transformer Note: Positive current direction is from the AC-system into the transformer
N_Tap_meas	ElmTr2	Reads the tap position and tapsize from the actual converter transformer
IchopPos_Meas	StaImea	Measures the chopper current at positive pole
IchopNeg_Meas	StaImea	Measures the chopper current at negative pole
HLC inputs	ElmDsl	These are the optional external controller inputs, but are not used
PWM converter	ElmVsc	Links the control frame outputs to the Voltage Source Converter (ElmVsc) used in the grid

Control	ElmDsl	Calls the external model library file (DLL)
INIT_WARN	ElmDsl	Detects and displays the initialization warnings
Chopper P	ElmValve	Links the Chopper activation signal (Gate signal) to the positive pole chopper element used in the grid
Chopper N	ElmValve	Links the Chopper activation signal (Gate signal) to the negative pole chopper element used in the grid
AC_Switch	ElmDsl	Trips the transformer if receives TripOrd = 1

3.3 Input values

Input values are the inputs to the control block. They are either measurement values or control signals that can be altered through user input (event change).

Table 5: Input values.

Signal name	Unit	Description
Time	s	Current time
tsPF	s	Simulation time step set in PowerFactory
UconvRe	kV	Measured value, Positive sequence converter bus voltage (real component)
UconvIm	kV	Measured value, Positive sequence converter bus voltage (imaginary component)
fRefMachine	pu	Frequency of the reference machine used as reference for the phase angle of UconvRe and UconvIm
UprimRe	kV	Measured value, Positive sequence PCC bus voltage (real component)
UprimIm	kV	Measured value, Positive sequence PCC bus voltage (imaginary component)
Udc_P	kV	Measured value, converter DC voltage, positive terminal
Udc_N	kV	Measured value, converter DC voltage, negative terminal
Idc_P	kA	Measured value, converter DC current, positive terminal
Idc_N	kA	Measured value, converter DC current, negative terminal
IvRe	kA	Measured value, Positive sequence converter bus current (real component)
IvIm	kA	Measured value, Positive sequence converter bus current (imaginary component)
IprimRe	kA	Measured value, PCCbus current (real component)
IprimIm	kA	Measured value, PCC bus current (imaginary component)
TapPosition	pu	Tap position
TapSize	%	Tap size
Ichop_P	kA	Measured value, chopper DC current, positive terminal
Ichop_N	kA	Measured value, chopper DC current, negative terminal
PModTotal	MW	Optional modulation signal for the Active power controller, not used
DeltaUacRef	kV	Optional modulation signal for the AC-voltage controller, not used

3.4 Parameter values

Parameter values are the parameters of the control model. These values can be changed in the PowerFactory ElmDsl parameter dialog, or by a PowerFactory Event at a certain time.

Table 6: Parameter values.

Parameter	Default	Min	Max	Unit	Description
ConvNr		1	99	-	Unique converter id number
PCtrlOn		0	1	-	1=PCtrl, 0=UdcCtrl
AcvCtrlOn		0	1	-	1=UacCtrl, 0=QCtrl
FvCtrlOn		0	1	-	1=FV control Mode, 0=Normal operation
ChopActive		0	1	-	1= Enable, 0= Disable
VsupportOn		0	1	-	1=Enable, 0=Disable
Deblock		0	1	-	1= Deblock, 0=Block
AsymmOp		0	1	-	0= Symmetric; 1= Asymmetric
RdcCable		0	99999	Ohm	DC Cable/Line resistance
UlineNom		1	99999	kV	Nominal voltage (zero tap) of transformer line side. ‘utrn_h‘ in the Converter transformer element should match this value
UvalveNom		1	99999	kV	Nominal voltage (zero tap) of transformer valve side. ‘utrn_l‘ in the Converter transformer element should match this value.
Sb_Trafo		1	99999	MVA	Base Power for the transformer. ‘strn‘ in the Converter transformer element should match this value
TrafoUk		0	100	%	Transformer reactance, in transformer base, in percent. ‘uktr‘ in the Converter transformer element should match this value
TapSide		1	2	-	Tap location, 1=Line, 2=Valve side
Lamr	53,94	0	99999	mH	Arm reactor inductance
IV_Max_Rec		0	99999	kA	Maximum valve current steady state in the rectifier
IV_Max_Inv		0	99999	kA	Maximum valve current steady state in the inverter
Q_Filter		-99999	99999	MVAr	If filter included in PQ-diagram, otherwise 0
Rchop		0.01	99999	Ohm	Chopper resistance, Total R if symmetrical DC system
EchopMax		0	99999	MJ	Maximum Total chopper energy
FundFreq		1	99999	Hz	Base Frequency
UlineBaseTool		1	99999	kV	Base voltage of PCC bus
Sb_Ctrl		1	99999	MVA	Base Power for the Control system of HVDC Light Converter. ‘Sbase’ in the PWM converter element should match this value.
UacBase		1	99999	kV	Base voltage for the control system of HVDC Light Converter. ‘Unom’ in the PWM converter element should match this value.

UdcNom		1	99999	kV	Nominal DC voltage Terminal-to-terminal. The 'Unomdc' in the PWM converter element should match this value.
UacDroop		0	100	%	UacDroop parameter
KgFV	0,5	0	99999	-	FV_KG, gain parameter when in FV control
Qmax		-99999	99999	MVAr	Converter maximum reactive power from PQ diagram
Qmin		-99999	99999	MVAr	Converter minimum reactive power from PQ diagram
IV_Trans_Max		0	99999	kA	Maximum transient current
TactivateDeltaF	9999	0	99999	s	Time constant to reset initial frequency offset in Passive Net mode
CctrlTf	0,002	0	99999	s	Time constant for LP-filter current ctrl
PLL_Kp	2,5	0	99999		Gain in PLL
PLL_Ti	0,1	0	99999	s	Integral Time constant in PLL
PrampUp	10	0	99999	pu/s	Max pos ramp rate of P ctrl
PrampDo	10000	0	99999	pu/s	Max neg ramp rate of P ctrl
QKp	0,5	0	99999	-	Gain reactive power control
Qti	0,04	0	99999	s	Integral time constant Q ctrl
UacKp	1	0	99999	-	Gain AC V control
UacTi	0,05	0	99999	s	Integral time constant AC V ctrl
UdcKp	4	0	99999	-	Gain in DC V ctrl
UdcTi	0,1	0	99999	s	Integral time constant DC V ctrl
ColKpO	15	0	99999	-	Gain DCCOL overvoltage
ColKpU	15	0	99999	-	Gain DCCOL undervoltage
UdColMx	1,05	0	99999	pu	DCCOL overvoltage activation
UdColMn	0,95	0	99999	pu	DCCOL undervoltage activation
IslFreq	0	-99999	99999	pu	Islanded f dev from from nom
PFNom	0	-99999	99999	pu	P for nominal f in FvCtrl
VsupKp	2	0	99999	-	Gain in V_sup at low voltage
VsupKpHi	5	0	99999	-	Gain in V_sup at high voltage
VsupLimH	1,1	0	99999	pu	Upper activation level for V_support
VsupLimL	0,85	0	99999	pu	Lower actiovation level for V_support
VsupSbPu	0,9	0	99999	pu	Part of Sb avaible for V_support
VsupHyst	0,03	0	99999	pu	Hystersis voltage support
VsupD	0	0	99999	pu	Deadband voltage support
ChopLvl	1,16	0	99999	pu	DC voltage chopper activation level
ChopHys	0,13	0	99999	pu	Chopper deactivation hystersis
P_REF		-99999	99999	MW	Active Power reference, use to change reference during simulation
Q_REF		-99999	99999	MVAr	Reactive power reference, use to change reference during simulation
UAC_REF		99999	99999	kV	AC voltage reference, use to change reference during simulation

UDC_REF	99999	99999	kV	DC voltage reference, use to change reference during simulation

3.5 Output values

Output values are used for traditional output, i.e., to be used to control voltage source in the electrical system (converter representation) or for plotting purposes.

Table 7: Output values.

Parameter	Unit	Description
Pmr	pu	Modulation index, real part
Pmi	pu	Modulation index, imaginary part
ChopOn	-	Chopper activation signal
InitWarn	-	Activate initialization warning
ModIndex	pu	Modulation index, amplitude
TripOrd	-	Trip order signal to the transformer
ModIndex	pu	Modulation index, use for plotting
Pline	pu	Control system internal filtered measured active power, use for plotting
Qline	pu	Control system internal filtered measured reactive power, use for plotting
Echopp	MJ	Estimated energy in chopper, use for plotting

3.6 Error message

The control system prints error messages to a PowerFactory output window. It is of big importance to check this when starting a simulation, to make sure that the model is setup correctly.

Error messages are also printed to a file on the system path for temporary files. The system path is defined by the system variable “temp” which can be shown by the windows command:
echo %temp%

This command can be executed in a windows command window, started with typing cmd in the windows search window. Typically this path is:

- C:\Users\AABBCC\AppData\Local\Temp, where AABBCC is the users name
- C:\Temp
- If not, %temp% is defined on the PowerFactory installation path

The used path will also be printed in the PowerFactory interface at initialization if the parameter Inspect is set to 1, refer to Section 3.4.
The file printed is:

- PowerFactory_Hitachi_HVDC_HVDCLIGHT1_Error.txt

It is important that the user have write permission to the used system path for temporary files, and that the path exist.

3.7 HVDC Light model Interface

The interface in PowerFactory to the external DLL is described in this section. The relationship between different elements in the HVDC Light model is shown in Figure 3.

There are different measuring objects and objects with inputs that is connected to the control block. The control block calls the DLL which contains the HVDC control system. The output signals from the control system, which are the real and imaginary parts of the modulation index (i.e., Pmr and Pmi), are then be passed on to the block named PWM converter, which is linked to the Voltage Source Converter used in the grid.

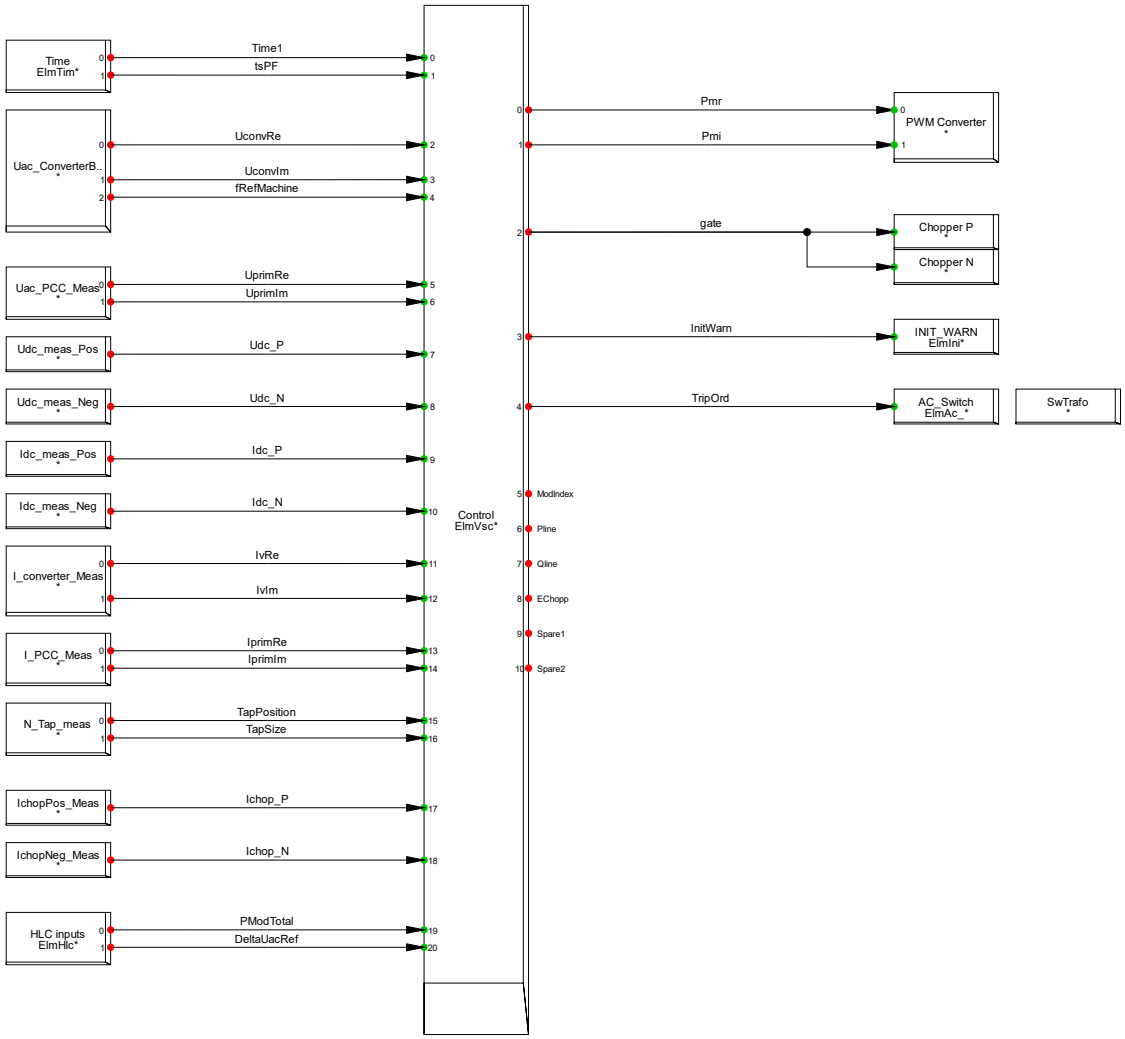


Figure 3: Overview of the control within PowerFactory.

It can be noted that the input fRefMachine is taken from the converter bus voltage measurement, the reference for the PLL in the converter control. The fRefMachine is used in the converter model PLL to properly pick up frequency deviations in the system.

Measurements, Control, Chopper and PWM Converter slots in Figure 3 are declared and referenced to the specific object in the Composite model shown in Figure 4.

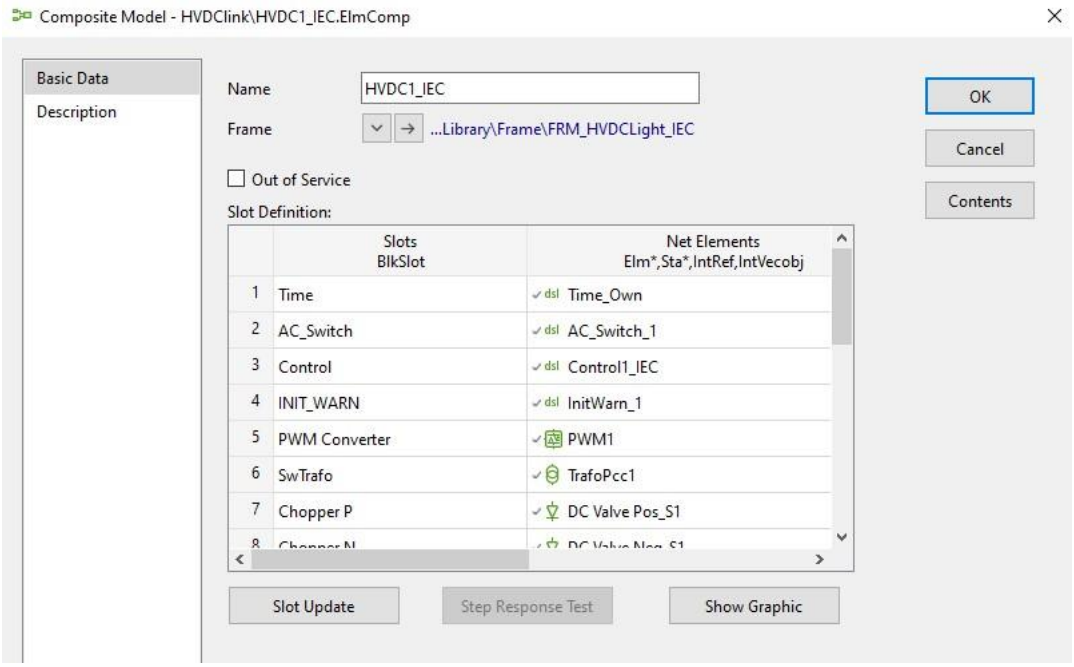


Figure 4: Control input interface.

Figure 5 shows the input parameters for the HVDC control system.

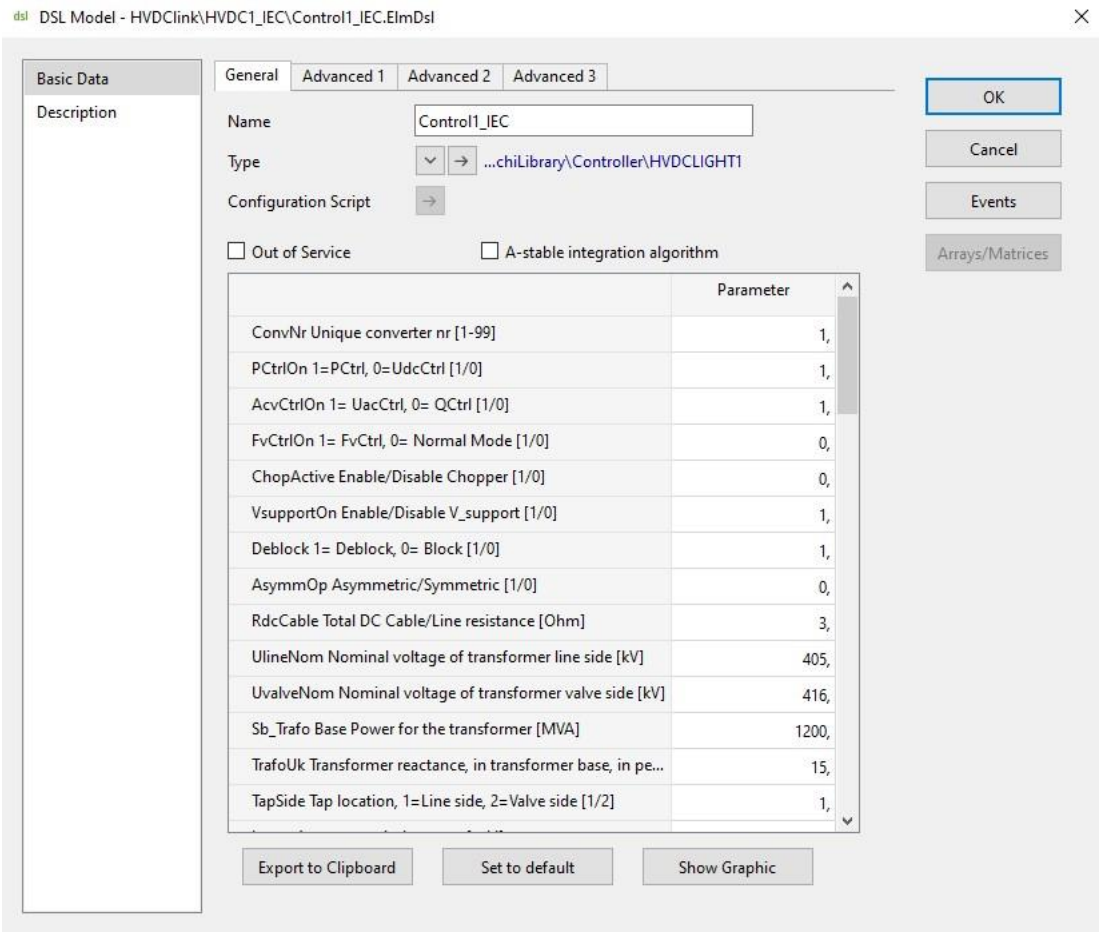


Figure 5: Parameter Interface of the control system.

The call to the external DLL is created in the Block Definition of the Common Model shown in Figure 6.

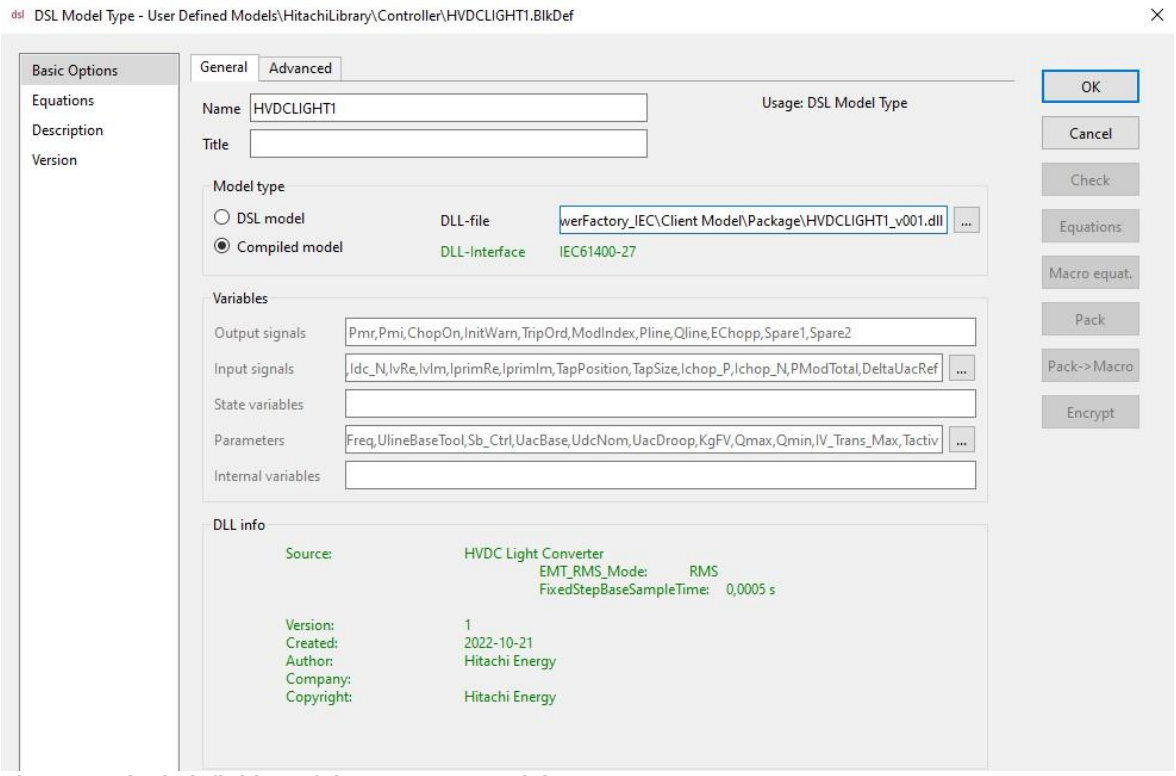


Figure 6: Block definition of the Common Model.

In this Block Definition, input signal and output signals are transferred between PowerFactory and the DLL.

3.8 Control system

The principles of the converter control represented in the model are given in Figure 7. In normal operation the user by can select one of the following for each station:

- AC voltage control
- Reactive power control

In the same way the user can choose one of the following for each station:

- DC voltage control
- Active power control

For AC voltage control and Reactive power control the operating modes can be chosen indecently of each other. For DC voltage control and Active power control one station must be in DC voltage control and the other in Active power control.

For an application where the HVDC is connected to an islanded network the HVDC can be set to work in Islanded Network Control mode. In this mode the HVDC will control the frequency and AC voltage of the AC network. In this setup, the other station must be set to DC voltage control mode.

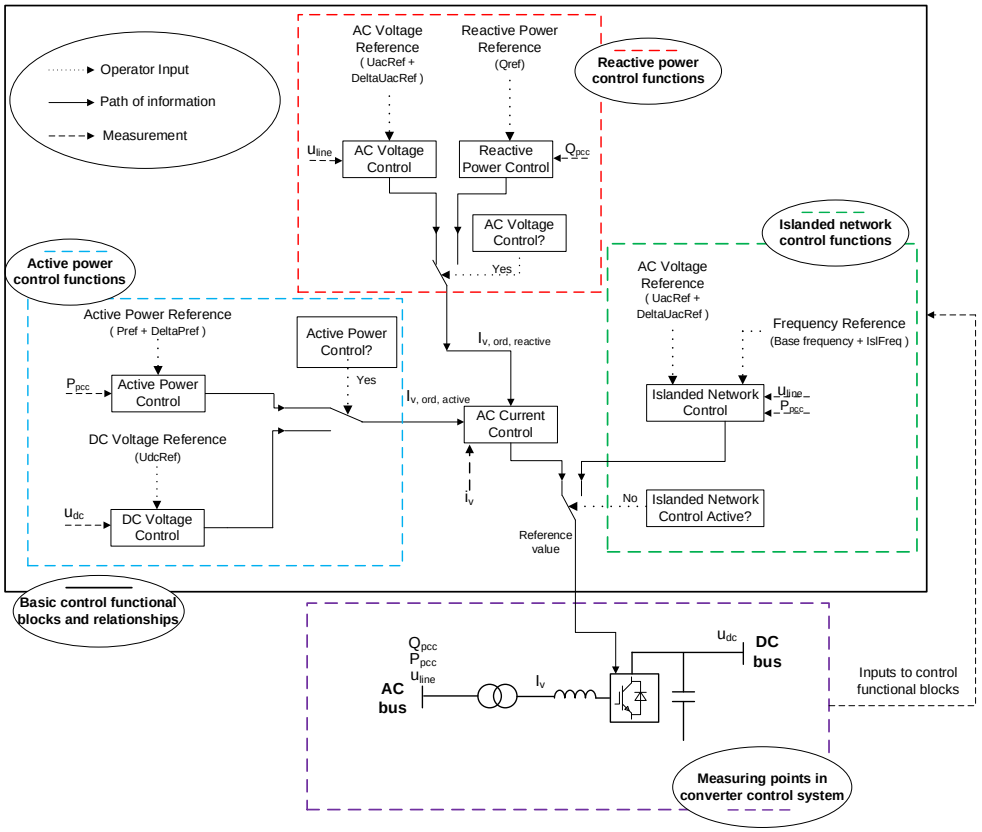


Figure 7: Principles of converter control.

More detailed information about HVDC Light control and the different control modes can be found in online document “HVDC Light It’s time to connect” [1].

4. References

[1] HVDC Light It’s time to connect," [Online]. Available: hitachienergy.com/hvdc

Hitachi Energy
Uno Lamm HVDC Center SE-771 80
Ludvika Sweden
Tel: +46 21 32 50 00

hitachienergy.com/hvdc

©Hitachi Energy 2024. All rights reserved.
Specifications subject to change without notice.

1JNL1270519 REV B